THE CANONICAL MORPHISM TO THE SPECTRUM OF GLOBAL FUNCTIONS IS DOMINANT OVER QUASI-COMPACT SCHEMES

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Editorial note. Counted span is the author's own proof environment, source lines 23577-23596, immediately following the statement at 23565-23575.

Editorial note. Two cross-chapter pointers (Morphisms, Schemes) are recorded in assembly.external_references.

Editorial note. The statement's three assertions (vanishing of the largest quasi-coherent submodule, injectivity, dominance) are all addressed inside the span.

Allowed standard prerequisites (editorial).

- The canonical morphism into the spectrum of global sections. Schemes, lemma schemes-lemma-morphism-into-affine, cited in the statement. Fixes the morphism f : X -> S = Spec(Gamma(X, O_X)) whose injectivity is studied.
- Scheme-theoretic image of a quasi-compact morphism. Morphisms, lemma morphisms-lemma-quasi-compact-scheme-theoretic-image, cited in the proof. Identifies the closed subscheme cutting out the kernel, which is then shown to be empty.
- Quasi-coherent modules on an affine scheme. used to pass from the kernel sheaf to an ideal of the ring of global functions. Turns the kernel of f^sharp into an ideal, so that vanishing is visible on the quasi-compact pieces.

Source reference. Original reference 01R8: focused-seeds/geometry/S9-morph-aff-injective-on-sheaves/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I1: focused-seeds/geometry/S9-morph-aff-injective-on-sheaves/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $f : X \to S = \mathrm{Spec}(\Gamma(X, \mathcal{O}_X))$ be the canonical morphism of Schemes, Lemma 01I1. The largest quasi-coherent $\mathcal{O}_S$-module contained in the kernel of $f^\sharp : \mathcal{O}_S \to f_*\mathcal{O}_X$ is zero. If X is quasi-compact, then $f^\sharp$ is injective. In particular, if X is quasi-compact, then f is a dominant morphism.*

Proof. Let $M \subset \Gamma(S, \mathcal{O}_S)$ be the submodule corresponding to the largest quasi-coherent $\mathcal{O}_S$-module contained in the kernel of $f^\sharp$. Then any element $a \in M$ is mapped to zero by $f^\sharp$. However, $f^\sharp(a)$ is the element of

$$\Gamma(S, f_*\mathcal{O}_X) = \Gamma(X, \mathcal{O}_X) = \Gamma(S, \mathcal{O}_S)$$

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corresponding to a itself! Thus $a = 0$. Hence $M = 0$ which proves the first assertion. Note that this is equivalent to the morphism $f : X \to S$ being scheme-theoretically surjective.

If X is quasi-compact, then $\mathrm{Ker}(f^\sharp)$ is quasi-coherent by Morphisms, Lemma 01R8. Hence $\mathrm{Ker}(f^\sharp) = 0$ and $f^\sharp$ is injective. In this case, f is a dominant morphism by part (4) of Morphisms, Lemma 01R8. $\square$